SEAT No.________

No. of Printed Pages : 2

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SARDAR PATEL UNIVERSITY

BCA Semester – 1 (Reg & NC)

Wednesday, **Date:** 14-8-2019

Session: Morning **Time: 10:00 A.M. to 1:00 P.M.**

Subject Code: US01CBCA01 **Total Marks: 70**

Subject Title : Fundamentals of Computer Programming Using 'C'

Q1. Multiple Choice Questions. (Attempt all) **[10]**

1. What symbol is used to represent output in a flowchart?
   - A. Rectangle C. Arrow
   - B. Parallelogram D. Circle
2. A translator is a program which translates instruction of ____________________.
   - A. High level language into Assembly language.
   - B. Machine language into High language.
   - C. Assembly into High language.
   - D. High level language or Assembly into Machine language.
3. _______ is format specifier to read and write character values.
   - A. %d C. %f
   - B. %c D. %ld
4. Which of the following escape sequence character constant is used for printing on a new line ?
   - A. \n C. \t
   - B. \b D. \c
5. _______ statement terminates the execution of loop containing it.
   - A. loop C. terminate
   - B. break D. end
6. Specify header file for utilizing mathematical functions.
   - A. conio C. string
   - B. math D. stdio
7. Two dimensional array requires ________ number of subscripts to identify array elements.
   - A. 0 C. 1
   - B. 2 D. 3
8. Strings ends with ____________.
   - A. '\n' C. '\0'
   - B. '\t' D. '\b'
9. Function parameters are also known as __________.
   - A. arguments C. values
   - B. list D. datatypes
10. When the function return any value at that time it return ______value per function call.
    - A. One C. Zero
    - B. Two D. Multiple

Q2. Answer the following short questions (Attempt any TEN) [20]

1. State advantages of High level languages over Machine level language.
2. What is an Editor? Give 3 examples of well know editors.
3. List all translators. Explain Assembler in brief.
4. List basic data types used in 'c' language.
5. Explain Basic structure of 'c' program.
6. Explain if statement with example in brief.
7. Write syntax to initialize 1D array in 'c'. Give two examples.
8. List out any four mathematical functions.
9. Explain isdigit() function with example.
10. Define function? List out the categories of function.
11. List elements of a Function.
12. Differentiate between gets() and scanf().

Q3.a. Define algorithm. List Characteristics of an algorithm. Write an algorithm to find maximum of given three numbers. [05]

b. Explain Machine level languages in detail. [05]

OR

Q3.a. Explain symbols used to draw flowchart. Write advantages and disadvantages of flowchart. [05]

b. Differentiate between Compiler and Interpreter. [05]

Q4.a. Explain Arithmetic, Relational and Increment-Decrement operators with example. [06]

b. Explain printf() and scanf() statement with syntax and example in brief. [04]

OR

Q4.a. Explain if..else and nested if statements with syntax and example. [06]

b. What is Variable? List rules to declare valid variable names. Also give example of valid variable names. [04]

Q5.a. Differentiate between : [06]

i) while and do..while **ii) break and continue**

Q5.b. Define Array? Explain syntax of declaration of One Dimensional Array and Two Dimensional Array with example. [04]

OR

Q5.a. Explain following functions with syntax , purpose and example: [06]

i) isupper() **ii) clrscr()** **iii) pow()**

Q5.b. Explain for loop statement in detail with example. [04]

Q6. Define String. Explain functions in detail with purpose, syntax and example. [10]

i) strrev() **ii) strlen()** **iii) strcpy()**

OR

Q6. Explain any two categories of user defined functions with example. [10]

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